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Fourier Series of Arbitrary Period $(c, c+2\pi)$

For funct^n having arbitrary length, for expansion of such funct^n in $[f(x), (c, c+2l)]$ fourier series, the transformation of variable as

Initially $f(x)$ length $2l$ $(c, c+2l)$

$$t = \frac{x\pi}{l} \Rightarrow x = \frac{tl}{\pi}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$\boxed{a_0 = \frac{1}{l}\int_c^{c+2l} f(x)\,dx}$$

$$\boxed{a_n = \frac{1}{l}\int_c^{c+2l} f(x)\cos\left(\frac{n\pi x}{l}\right)dx}$$

$$\boxed{b_n = \frac{1}{l}\int_c^{c+2l} f(x)\sin\left(\frac{n\pi x}{l}\right)dx}$$

Q. Expand $f(x) = +e^{-x}$ as a Fourier Series in interval $(-l, +l)$

Sol^n $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1} b_n \sin\left(\frac{n\pi x}{l}\right)$

$$a_0 = \frac{1}{l}\int_{-l}^{l} e^{-x}\,dx$$

$$a_0 = \frac{1}{l} - \left[e^{-x}\right]_{-l}^{l} = \frac{1}{l}\left[e^{-x}\right]_{l}^{-l} = \frac{1}{l}\left[e^{+l} - e^{-l}\right]$$

$$a_0 = \frac{2}{l}\,\frac{\left[e^{l} - e^{-l}\right]}{2}$$

$$a_0 = \frac{2}{l}\sinh l$$

$$a_n = \frac{1}{l}\int_{-l}^{l} e^{-x} \cos\left(\frac{n\pi x}{l}\right) dx$$

$$\cancel{\int uv\,dx = uv_1 - u'v_2 + u''v_3 - \ldots}$$

$$\cancel{a_n = \frac{1}{l}\; -\cos\left(\frac{n\pi x}{l}\right) e^{-x} + \frac{l}{n\pi}\sin\left(\frac{n\pi x}{l}\right) e^{-x} + \cos\left(\frac{n\pi x}{l}\right)}$$

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)$$

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)$$

$$\sinh x = \frac{e^{x} - e^{-x}}{2}$$

$$\cosh x = \frac{e^{x} + e^{-x}}{2}$$

$$\cancel{a_n = \frac{1}{}} \quad a = -1,\ b = \frac{n\pi}{l}$$

$$a_n = \frac{1}{l}\left[\frac{e^{-x}}{1+\frac{n^2\pi^2}{l^2}}\left[-\cos\left(\frac{n\pi}{l}\right)x + \frac{n\pi}{l}\sin\left(\frac{n\pi}{l}\right)x\right]\right]_{-l}^{l}$$

$$a_n = \frac{l^{\cancel{2}}}{\cancel{l}(l^2+n^2\pi^2)}\left[-e^{-x}\cos\left(\frac{n\pi}{l}\right)x + \frac{n\pi}{l}\sin\left(\frac{n\pi}{l}\right)x\right]_{-l}^{l}$$

$$a_n = \frac{+l}{l^2+n^2\pi^2}\left[-e^{-l}\cos(n\pi) + e^{+l}\cos n\pi\right]$$

$$a_n = \frac{l}{l^2+n^2\pi^2}\left[-e^{-l}(-1)^n + e^{l}(-1)^n\right]$$

$$= \frac{(-1)^n\, l\, 2}{l^2+n^2\pi^2}\left(\frac{e^{l} - e^{-l}}{2}\right)$$

$$a_n = \frac{2l(-1)^n}{l^2+n^2\pi^2}\sinh l$$

$$b_n = \frac{1}{l}\int_{-l}^{l} e^{-x}\sin\left(\frac{n\pi x}{l}\right) dx$$

$$a = -1,\ b = \frac{n\pi}{l}$$

$$b_n = \frac{1}{l}\,\frac{e^{-x}}{1+\frac{n^2\pi^2}{l^2}}\left[-\sin\left(\frac{n\pi}{l}\right)x - \frac{n\pi}{l}\cos\left(\frac{n\pi}{l}\right)x\right]$$

$$b_n = \frac{-l}{l^2+n^2\pi^2}\left[e^{-x}\sin\left(\frac{n\pi}{l}\right)x + \frac{n\pi}{l}e^{-x}\cos\left(\frac{n\pi}{l}\right)x\right]_{-l}^{l}$$

$$b_n = \frac{-l}{l^2+n^2\pi^2}\left[\frac{n\pi}{l}e^{-l}\cos(n\pi) - \frac{n\pi}{l}e^{l}\cos(n\pi)\right]$$

$$b_n = \frac{-\cancel{l}}{l^2+n^2\pi^2}\frac{n\pi}{\cancel{l}}\cos(n\pi)\left(e^{-l}-e^{l}\right)$$

$$= \frac{n\pi(-1)^n 2}{l^2+n^2\pi^2}\left[\frac{e^{l}-e^{-l}}{2}\right]$$

$$b_n = \frac{2n\pi(-1)^n}{l^2+n^2\pi^2}\sinh l$$

Half Range Fourier Series

$f(x) \Rightarrow (0,\pi)$

Extend from $(-\pi,0)$ $(0,\pi)$

① Cosine Series → To generate even funct^n, i.e, $f(x)=f(-x)$. Extend the funct^n is range $(-\pi,0)$, resultant half range cosine Fourier Series, $\boxed{b_n=0}$

$$\boxed{f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx}$$

$$\boxed{a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx}$$

$$\boxed{a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx}$$

② Sine Series → Extend the funct^n in interval $(-\pi,0)$. So that odd funct^n generates, i.e., $f(x)=-f(-x)$.

$$\boxed{a_0=a_n=0}$$

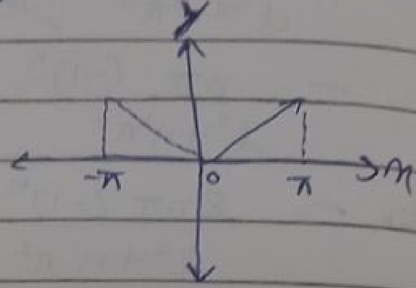
$$\boxed{f(x) = \sum_{n=1}^{\infty} b_n \sin nx}$$

$$\boxed{b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx}$$

Q. Obtain sine & cosine series for $f(x) = x$ the interval $0 \le x \le \pi$ and deduce

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots = ?$$

Sol^n Cosine series → extending funct^n in inter $(-\pi, 0)$ so that even function $f(x) = f(-x)$ is generated.



$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x\,dx$$

$$a_0 = \frac{\cancel{2}}{\cancel{\pi}} \frac{\pi^{\cancel{2}}}{\cancel{2}}$$

$$\boxed{a_0 = \pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx$$

$$= \frac{2}{\pi} \left[\frac{x}{n} \sin nx - \frac{1}{n} \int \sin nx \, dx \right]$$

$$= \frac{2}{\pi} \left[\frac{x}{n} \sin nx + \frac{1}{n^2} \cos nx \right]_0^{\pi}$$

$$\boxed{a_n = \frac{2}{\pi n^2} \left[(-1)^n - 1 \right]}$$

$$a_n = \left\{ \begin{array}{ll} 0, & n \text{ is even} \\ \frac{-4}{\pi n^2}, & n \text{ is odd} \end{array} \right\}$$

$$x = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left(\frac{-4}{\pi n^2} \right) \cos nx$$

$$x = \frac{\pi}{2} - \frac{4}{\pi} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots \right.$$

Put $x = 0$

$$0 = \frac{\pi}{2} - \frac{4}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots \right]$$

$$+\frac{\pi}{2} \times \left(+\frac{\pi}{4}\right) = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots$$

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$$

Q. $f(x) =$

Q. $f(x) = \begin{cases} x & ; 0 < x < \frac{\pi}{2} \\ \pi - x & ; \frac{\pi}{2} < x < \pi \end{cases}$ Show that

i) $f(x) = \frac{4}{\pi}\left[\sin x - \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} + \dots\right]$

ii) $f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left[\frac{\cos 2x}{1^2} + \frac{\cos 6x}{3^2} + \frac{\cos 10x}{5^2} + \dots\right]$

Sol^n sin Series

$a_0 = a_n = 0$

$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$

$b_n = \frac{2}{\pi}\int_0^{\pi} f(x) \sin nx \, dx$

$= \frac{2}{\pi}\left[\int_0^{\frac{\pi}{2}} f_1(x) \sin nx \, dx + \int_{\frac{\pi}{2}}^{\pi} f_2(x) \sin nx \, dx\right]$

$b_n = \frac{2}{\pi}\left[\int_0^{\frac{\pi}{2}} x \sin nx \, dx + \int_{\frac{\pi}{2}}^{\pi} \pi \sin nx \, dx - \int_{\frac{\pi}{2}}^{\pi} x \sin nx \, dx\right]$

$\int \underset{u}{x} \underset{v}{\sin nx} \, dx = \frac{-x \cos nx}{n} + \frac{1}{n}\int \cos nx \, dx$

$\int x \sin nx \, dx = -\frac{x}{n}\cos nx + \frac{\sin nx}{n^2}$

$b_n = \frac{2}{\pi}\left\{\left[-\frac{x}{n}\cos nx + \frac{\sin nx}{n^2}\right]_0^{\frac{\pi}{2}} - \frac{\pi}{n}\left[\cos nx\right]_{\frac{\pi}{2}}^{\pi} - \left[\frac{-x \cos nx}{n} + \frac{\sin nx}{n^2}\right]\right.$

$b_n = \frac{2}{\pi}\left\{\left[\frac{-\pi}{2n}\cos\frac{n\pi}{2} + \frac{1}{n^2}\sin\frac{n\pi}{2}\right] - \frac{\pi}{n}\left(\cos n\pi - \cos\frac{n\pi}{2}\right)\right.$

$+ \left[\left(-\frac{\pi}{n}\cos n\pi + 0\right) + \left(\frac{-\pi}{2n}\cos\frac{n\pi}{2} + \frac{\sin\frac{n\pi}{2}}{n^2}\right)\right]\Bigg\}$

$b_n = \frac{2}{\pi}\left[\frac{-\pi}{2n}\cos\frac{n\pi}{2} + \frac{1}{n^2}\sin\frac{n\pi}{2} \cancel{- \frac{\pi}{n}\cos n\pi} + \frac{\pi}{n}\cos\frac{n\pi}{2}\right.$

$\left.\cancel{\frac{\pi}{n}\cos n\pi} - \frac{\pi}{2n}\cos n\pi + \frac{\sin\frac{n\pi}{2}}{n^2}\right]$

$$b_n = \frac{4}{\pi n^2} \sin\frac{n\pi}{2}$$

$$b_n = \begin{cases} \frac{4}{\pi n^2} \sin\frac{n\pi}{2}, & \text{when } n \text{ is odd} \\ 0, & \text{when } n \text{ is even} \end{cases}$$

$$f(x) = \sum_{\substack{n=1 \\ (\text{odd})}}^{\infty} b_n \sin nx$$

$$= \sum_{\substack{n=1 \\ (\text{odd})}}^{\infty} \frac{4}{\pi n^2} \sin\left(\frac{n\pi}{2}\right) \sin nx$$

$$= \frac{4}{\pi} \left[\frac{1}{1^2} \sin\frac{\pi}{2} \sin x + \frac{1}{3^2} \sin\frac{3\pi}{2} \sin 3x + \frac{1}{5^2} \sin\frac{5\pi}{2} \sin 5x + \dots \right]$$

(i) $$f(x) = \frac{4}{\pi} \left[\frac{1}{1^2} \sin x - \frac{1}{3^2} \sin 3x + \frac{1}{5^2} \sin 5x + \dots \right]$$

H.P.

X

(ii) Cosine series → extending the interval in $(-\pi, 0)$ so that even funct^n is generated.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{2}{\pi} \left[\int_0^{\frac{\pi}{2}} x \cancel{\cos nx}\, dx + \int_{\frac{\pi}{2}}^{\pi} (\pi - x) \cancel{\cos nx}\, dx \right]$$

$$= \frac{2}{\pi} \left[\left(\frac{\pi}{2}\right)^2 + \pi\left(\pi - \frac{\pi}{2}\right) - \frac{1}{2}\left(\pi^2 - \frac{\pi^2}{4}\right) \right]$$

$$= \frac{2}{\pi} \left[\frac{\pi^2}{4} + \cancel{\frac{\pi^2}{2}} - \cancel{\frac{\pi^2}{2}} + \frac{\pi^2}{4} \right]$$

$$= \frac{\cancel{2}}{\cancel{\pi}} \times \frac{\pi^{\cancel{2}}}{\cancel{4}\, 2}$$

$$a_0 = \frac{\pi}{2}$$

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$$a_n = \frac{2}{\pi}\left[\int_0^{\frac{\pi}{2}} x\cos nx\,dx + \int_{\frac{\pi}{2}}^{\pi} (\pi - x)\cos nx\,dx\right]$$

Develop $\sin\left(\frac{\pi x}{l}\right)$ in half range cosine series in range $0<x<l$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

$$a_n = \frac{2}{l}\int_0^l \sin\left(\frac{\pi x}{l}\right) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$a_n = \frac{1}{l}\int_0^l \sin (n+1)\frac{\pi x}{l} - \sin(n-1)\frac{\pi x}{l}\, dx$$

$$a_n = \frac{1}{l}\left[\frac{-l}{\pi(n+1)}\cos (n+1)\frac{\pi x}{l} + \frac{l}{\pi(n-1)}\cos(n-1)\frac{\pi x}{l}\right]_0^l$$

$$= \frac{1}{l}\left[-\frac{l}{\pi}\times\frac{1}{n+1}\cos(n+1)\pi + \frac{l}{\pi(n-1)}\cos(n-1)\pi\right]$$

$$\cancel{a_n = \frac{1}{l}\frac{l}{\pi}\left[\frac{1}{n-1}\cos(n-1)\pi - \frac{1}{n+1}\cos(n+1)\pi\right]}$$

$$a_0 = \frac{2}{l}\int_0^l \sin\frac{\pi x}{l}\, dx$$

$$= \frac{-2}{\cancel{l}}\times\frac{\cancel{l}}{\pi}\left[\cos\frac{\pi x}{l}\right]_0^l$$

$$= -\frac{2}{\pi}\left[\cos\pi - \cos 0\right]$$

$$a_0 = \frac{+4}{\pi}$$

$$a_n = \frac{1}{l}\left[\frac{-l}{\pi(n+1)}\cos(n+1)\pi + \frac{l}{\pi(n+1)} + \frac{l}{\pi(n-1)}\cos(n-1)\pi - \frac{l}{\pi(n-1)}\right]$$

$$= \frac{1}{\cancel{l}}\times\cancel{l}\left[\frac{1}{\pi(n+1)}\left(1-\cos(n+1)\pi\right) + \frac{1}{\pi(n-1)}\left(\cos(n-1)-1\right)\right]$$

$$a_n = \frac{1}{\pi}\left[\frac{1-\cos(n+1)\pi}{(n+1)} + \frac{\cos(n-1)\pi - 1}{(n-1)}\right]$$

$$a_n = \begin{cases} 0, & n \text{ is odd, } n\neq 1 \\ \dfrac{-4}{\pi(n+1)(n-1)}, & n \text{ is even} \end{cases}$$

then, $$a_n = \frac{1}{\pi}\left[\frac{1-\cos 3\pi}{(n+1)} + \frac{\cos \pi - 1}{n-1}\right]$$

$$= \frac{1}{\pi}\left[\frac{2}{n+1} - \frac{2}{n-1}\right]$$

$$= \frac{2}{\pi}\,\frac{(n-1)-(n+1)}{(n+1)(n-1)}$$

$$a_n = \frac{-4}{\pi(n+1)(n-1)}$$

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FOURIER INTEGRAL THEOREM

Let $f(x)$ is an integral satisfying Drichlet's condition in a finite interval $-l<x<l$ and represents as $\frac{1}{2}[f(x-0)+f(x+0)]$ defined at every pt. of discontinuity. Let integral $\int_{-\infty}^{\infty}|f(x)|dx$ converges, i.e., $f(x)$ is integrable at every pt. in limit $-\infty<x<\infty$. Then,

$$\boxed{f(x)=\frac{1}{\pi}\int_0^\infty\int_{-\infty}^{\infty} f(t)\cos(t-x)\,dt\,dw}$$

(i) Sine Series $\rightarrow$

$$\boxed{f(x)=\frac{2}{\pi}\int_0^\infty \sin wx\,dw\int_0^\infty f(t)\sin wt\,dt}$$

(ii) Cosine Series $\rightarrow$

$$\boxed{f(x)=\frac{2}{\pi}\int_0^\infty \cos wx\,dw\int_0^\infty f(t)\cos wt\,dt}$$

dw वाले को solve नहीं करेंगे

Q. Express the funct^n

$$f(x)=\begin{cases}1 & \text{when } |x|\le 1\\ 0 & \text{when } |x|>1\end{cases}$$ as a Fourier integr

& hence evaluate $\int_0^\infty \frac{\sin\lambda\cos\lambda x}{\lambda}\,d\lambda$

Sol^n $f(x)=1$, $f(-x)=1$

$f(x)=f(-x) \Rightarrow$ Even

Cosine Fourier Integral

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$$f(x) = \frac{2}{\pi}\int_0^\infty \cos x\lambda \, d\lambda \int_0^\infty f(t) \cos \lambda t \, dt$$

$|x| \leq 1$ | $|x| > 1$

$-1 \leq x \leq 1$ | $x > 1 \cap x < -1$

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos x\lambda \, d\lambda \left[\int_0^1 f(t) \cos \lambda t \, dt + \int_1^\infty f(t) \cos \lambda t \, dt\right]$$

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos x\lambda \, d\lambda \left[\int_0^1 (1) \cos \lambda t \, dt + 0\right]$$

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos x\lambda \, d\lambda \left[\frac{\sin \lambda t}{\lambda}\right]_0^1$$

$$f(t) = \frac{2}{\pi}\int_0^\infty \frac{\sin \lambda \cos \lambda x}{\lambda} \, d\lambda$$

$$f(t) \cdot \frac{\pi}{2} = \int_0^\infty \frac{\sin \lambda \cos \lambda x}{\lambda} \, d\lambda = \begin{cases} \frac{\pi}{2}, & \text{when } |x| \leq 1 \\ 0, & \text{when } |x| > 1 \end{cases}$$

Clearly, $f(x)$ is discontinuous at $x = 1$

$$f(1) = \frac{f(x-0) + f(x+0)}{2} = \frac{\frac{\pi}{2} + 0}{2} = \boxed{\frac{\pi}{4}}$$

FOURIER COMPLEX INTEGRAL

$$\boxed{f(x) = \frac{1}{2\pi}\int_{-\infty}^\infty e^{-i\omega x} \, d\omega \int_{-\infty}^\infty f(t) \, e^{i\omega t} \, dt}$$

Q. Express $f(x) = \begin{cases} 1 & \text{for } 0 \le x \le \pi \\ 0 & \text{for } x > \pi \end{cases}$ as Fourier sine integral.

Evaluate $\int_0^\infty \frac{1-\cos\pi\lambda}{\lambda} \sin\lambda x \, d\lambda$

Solⁿ Sine Series

$$f(x) = \frac{2}{\pi}\int_0^\infty \sin wx \, dw \int_0^\infty f(t) \sin wt \, dt$$

$$f(x) = \frac{2}{\pi}\int_0^\infty \sin\lambda x \, d\lambda \int_0^\pi \sin\lambda t \, dt$$

$$= \frac{2}{\pi}\int_0^\infty \sin\lambda x \, d\lambda \left[-\frac{\cos\lambda t}{\lambda}\right]_0^\pi$$

$$= -\frac{2}{\pi}\int_0^\infty \frac{\sin\lambda x \, d\lambda}{\lambda}\left[\cos\lambda\pi - 1\right]$$

$$f(x) = \frac{2}{\pi}\int_0^\infty \sin\lambda x \, d\lambda \left(\frac{1-\cos\pi\lambda}{\lambda}\right)$$

$$f(x)\frac{\pi}{2} = \int_0^\infty \sin\lambda x \frac{(1-\cos\pi\lambda)}{\lambda} \, d\lambda = \begin{cases} \frac{\pi}{2} & \text{for } 0 \le x \le \pi \\ 0 & \text{for } x > \pi \end{cases}$$

$f(x)$ discont. at $x = \pi$

$$f(\pi) = \frac{f(\pi-0)+f(\pi+0)}{2} = \frac{\frac{\pi}{2}+0}{2} = \frac{\pi}{4}$$

Q. Find complex form of Fourier integral expression

$$f(x) = \begin{cases} e^{-kx}, & x > 0, \ k > 0 \\ 0, & \text{otherwise} \end{cases}$$

Ans $\frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{e^{-i\omega x}}{k - i\omega} \, d\omega$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega x} \, d\omega \int_{-\infty}^{\infty} f(t) \, e^{i\omega t} \, dt$$